

Software Engineering

Requirements Engineering

What are Requirements in Software Engineering?

Non-functional requirements (NFR)

- Describe the quality of a system

Use quality criteria to align on the quality scenarios for your system

Make sure non-functional requirements are measurable

(e.g. “within 3 seconds”, “for 500k users”)

Make deliberate tradeoffs between different quality criteria

(“e.g. we trade usability for security at the login screen”)

Functional Requirements (FR)

- Describe the features of the system

Create user stories to align on the features of your system from the user’s perspective

Functional Requirements

- Requirements Engineering is an iterative process
- Requirements will be
 - refined – fleshed out details for development
 - changed – modified to better fit the customer need
 - removed – parts of the requirement might get obsolete once details are clear
 - thrown away completely – the entire requirement might get obsolete
- Be careful not to produce “waste”: only specify (in detail) what you need to build soon

Defining Requirements: User Stories

User Stories (Examples)

Create account:

As administrator

I want to create a new account for a *customer* with their *email, first name and last name* so that they can get access to the system.

Get access:

As customer

I want to login to the system so that I have personalised access to my data.

Search account:

As administrator

I want to search for customers by *email, first name or last name* so that I can manage them.

Delete account:

As administrator

I want to delete a *customer* account so that I can revoke access to the system.

The User Story Template

Goal: Define Requirements

As [ROLE]

I want to [DO SOMETHING]

so that I [ACHIEVE SOMETHING]

This template helps to

- Identify roles that use the product
- Understand the goals of the users
- Describe workflows

Important:

No technical details like “click on this button” to leave the design and implementation open.

Exercise: User Stories

Marketing created this pitch for the software you should create with your team:

MentorHive — Unlock Expertise Anytime, Anywhere

Finding the right mentor shouldn't be complicated. With MentorHive, you can instantly connect with professionals across every industry — from tech and design to finance and marketing — all ready to guide you with real-world experience.

MentorHive is the smart, streamlined platform that matches you with curated mentors based on your goals, skill level, and availability. Whether you're switching careers, leveling up your current role, or exploring a new passion, MentorHive makes expert guidance accessible in just a few clicks.

Exercise: User Stories

What makes MentorHive different for learners?

- Pair with mentors whose background truly fits your goals.
- Video, screen-sharing, and collaborative tools built right in.
- Set milestones and track your progress as you grow.

What makes MentorHive different for mentors?

- Share your knowledge with motivated learners.
- Take calls when it fits your schedule.
- Get compensated reliably without administrative work.
- See how your mentees grow with your guidance.

Start your mentorship journey today — and turn ambition into action.

Exercise: individually, write two user stories based on the software scenario.

Refining Requirements: Example Mapping

Example Mapping

Goal: Refine requirements

Collaboratively, i.e. together, the team develops for each user story

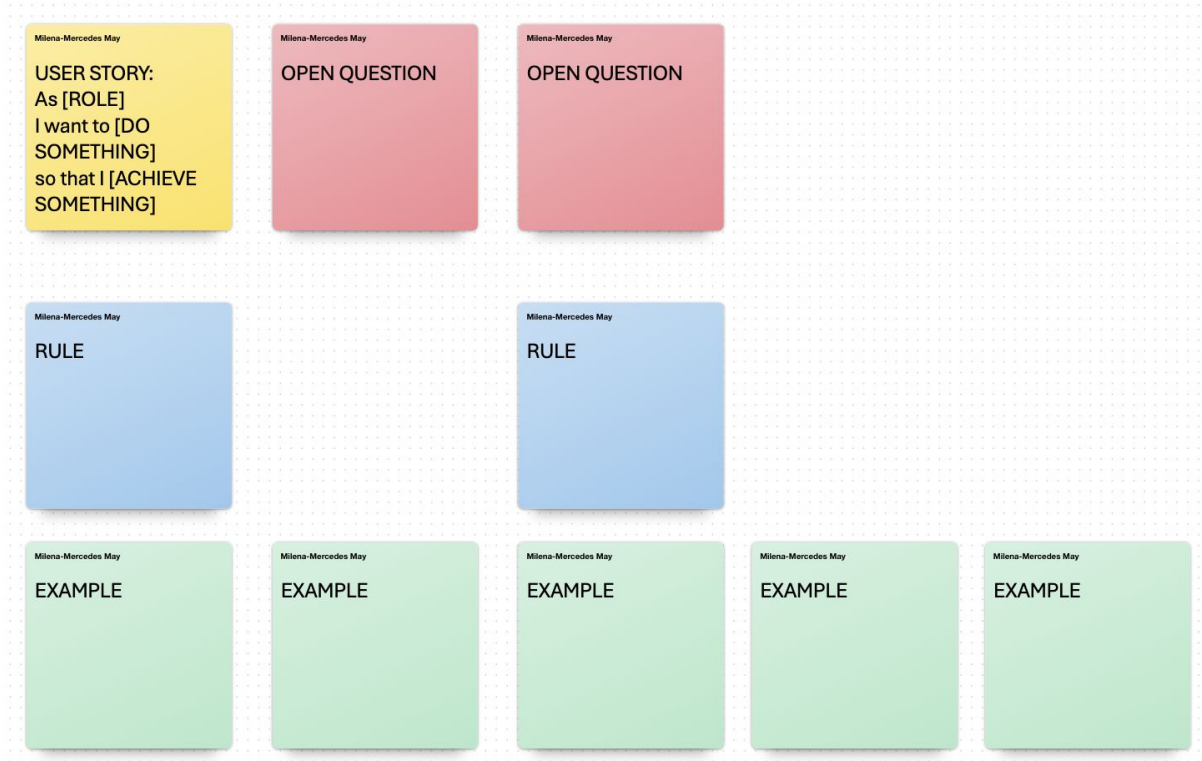
- Rules and for each rule
 - Examples that apply this rule

In doing so, they discover

- New user stories
 - possibly by splitting existing user stories
- Open questions
 - which must be clarified at the latest before the start of development

[Source: Cucumber Blog](#)

Example Mapping



Exercise: Example Mapping

Marketing created this pitch for the software you should create with your team:

MentorHive — Unlock Expertise Anytime, Anywhere

Finding the right mentor shouldn't be complicated. With MentorHive, you can instantly connect with professionals across every industry — from tech and design to finance and marketing — all ready to guide you with real-world experience.

MentorHive is the smart, streamlined platform that matches you with curated mentors based on your goals, skill level, and availability. Whether you're switching careers, leveling up your current role, or exploring a new passion, MentorHive makes expert guidance accessible in just a few clicks.

Exercise: Example Mapping

What makes MentorHive different for learners?

- Pair with mentors whose background truly fits your goals.
- Video, screen-sharing, and collaborative tools built right in.
- Set milestones and track your progress as you grow.

What makes MentorHive different for mentors?

- Share your knowledge with motivated learners.
- Take calls when it fits your schedule.
- Get compensated reliably without administrative work.
- See how your mentees grow with your guidance.

Start your mentorship journey today — and turn ambition into action.

Exercise: split into groups of 3 people and do an Example Mapping on a whiteboard tool, e.g. <https://whiteboard.cloud.microsoft/> for 20 minutes

Refining Requirements: Acceptance Criteria

Acceptance Criteria

Goal: Make requirements testable

Acceptance Criteria describe when a User Story is done

- Can be tested or verified - either passes or fails
- Focus on the outcome, not how the outcome is achieved
- As specific as possible, measurable if relevant

[Source: Scrum Alliance Blog](#)

Acceptance Criteria

Goal: Make requirements testable

Examples of Acceptance Criteria in bullet points

For a checkout page on a store's website:

- PayPal, Google Pay, Apple Pay, and all major credit cards can be used to complete the transaction
- User is prompted to log in if they aren't already

For an emailed water bill statement:

- Displays amount due
- Displays due date
- Displays possible late fee
- Allows user to log in and pay easily
- Lets user know how they can contact the office with questions or concerns

[Source: Scrum Alliance Blog](#)

Acceptance Criteria

Goal: Make requirements testable

Examples of Acceptance Criteria in Gherkin (Given - When - Then)

For an upcoming conference:

Given that I'm a registered attendee

When I enter the conference venue

Then branded signs will lead me to the registration table

For a health clinic's pre-appointment paperwork:

Given that I'm an existing health center patient,

When I schedule an appointment online,

Then I will receive the fillable pre-appointment paperwork electronically

[Source: Scrum Alliance Blog](#)

Exercise: Acceptance Criteria

Marketing created this pitch for the software you should create with your team:

MentorHive — Unlock Expertise Anytime, Anywhere

Finding the right mentor shouldn't be complicated. With MentorHive, you can instantly connect with professionals across every industry — from tech and design to finance and marketing — all ready to guide you with real-world experience.

MentorHive is the smart, streamlined platform that matches you with curated mentors based on your goals, skill level, and availability. Whether you're switching careers, leveling up your current role, or exploring a new passion, MentorHive makes expert guidance accessible in just a few clicks.

Exercise: Acceptance Criteria

What makes MentorHive different for learners?

- Pair with mentors whose background truly fits your goals.
- Video, screen-sharing, and collaborative tools built right in.
- Set milestones and track your progress as you grow.

What makes MentorHive different for mentors?

- Share your knowledge with motivated learners.
- Take calls when it fits your schedule.
- Get compensated reliably without administrative work.
- See how your mentees grow with your guidance.

Start your mentorship journey today — and turn ambition into action.

Exercise: as a group of 3 people, for two of your user stories, create three acceptance criteria.

